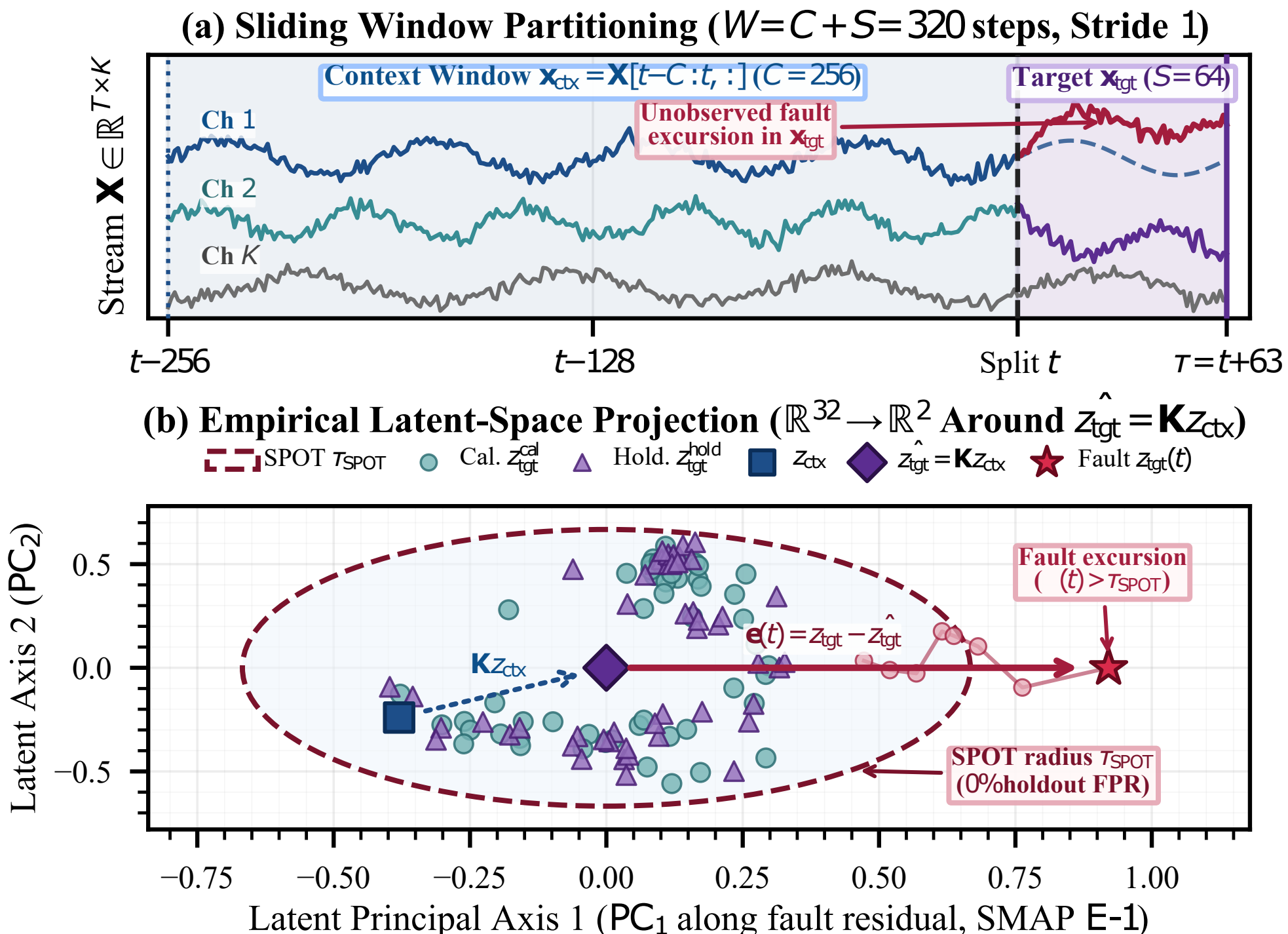

(a) Sliding Window Partitioning ($W = C + S = 320$ steps, Stride 1)
Stream $\mathbf{X} \in \mathbb{R}^{T \times K}$
Context Window $\mathbf{x}_{ctx} = \mathbf{X}[t-C:t, :]$ ($C = 256$)
Target $\mathbf{x}_{tgt}$ ($S = 64$)
Unobserved fault excursion in $\mathbf{x}_{tgt}$
Ch 1
Ch 2
Ch K
$t-256$
$t-128$
Split t
$\tau = t+63$
(b) Empirical Latent-Space Projection ($\mathbb{R}^{32} \to \mathbb{R}^2$ Around $\hat{z}_{tgt} = \mathbf{K} z_{ctx}$)
SPOT τ_{SPOT}
Cal. z_{tgt}^{cal}
Hold. z_{tgt}^{hold}
z_{ctx}
$\hat{z}_{tgt} = \mathbf{K} z_{ctx}$
Fault $z_{tgt}(t)$
Fault excursion ($(t) > \tau_{SPOT}$)
$\mathbf{e}(t) = z_{tgt} - \hat{z}_{tgt}$
$\mathbf{K} z_{ctx}$
SPOT radius τ_{SPOT} (0%holdout FPR)
Latent Axis 2 (PC_2)
0.5
0.0
−0.5
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Latent Principal Axis 1 (PC_1 along fault residual, SMAP E-1)